From Quantum Breakthroughs to Data Center-Ready Quantum Secure Networks

Remon Berrevoets

CTIO and Co-Founder, Falqon® Systems – Consortium Member QSNP

QSN PARTNERSHIP

Falqon Systems

OCP FUTURE TECHNOLOGIES SYMPOSIUM

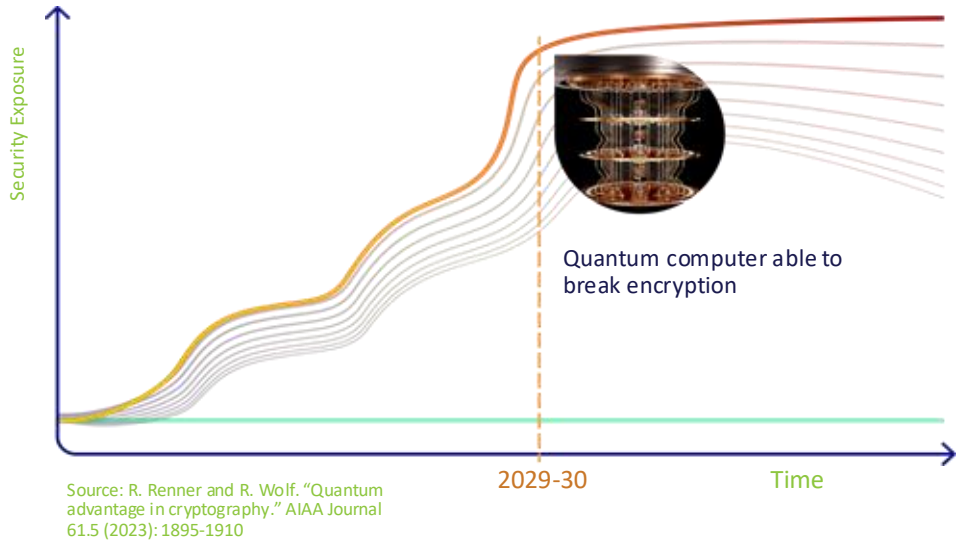
October 14, 2026
San Jose, California

Falqon Systems

The Quantum Threat Is Real And Its Already Started

Warning Signals

- Cybersecurity is already under growing pressure from AI enabled cyber threats
- Traditional encryption will collapse once quantum computers scale
- Encrypted data is already being harvested today
- Governments and banks are investing billions to prepare



Source: R. Renner and R. Wolf. "Quantum advantage in cryptography." AIAA Journal 61.5 (2023): 1895-1910

Security Risk & Impact

- Shut down of systems
- Disruption of operations
- Tampering of control signals
- Interception of critical data
- Harvest now-decrypt later of sensitive information

Securing communications with quantum technology is no longer optional – it's inevitable!

Falqon Systems

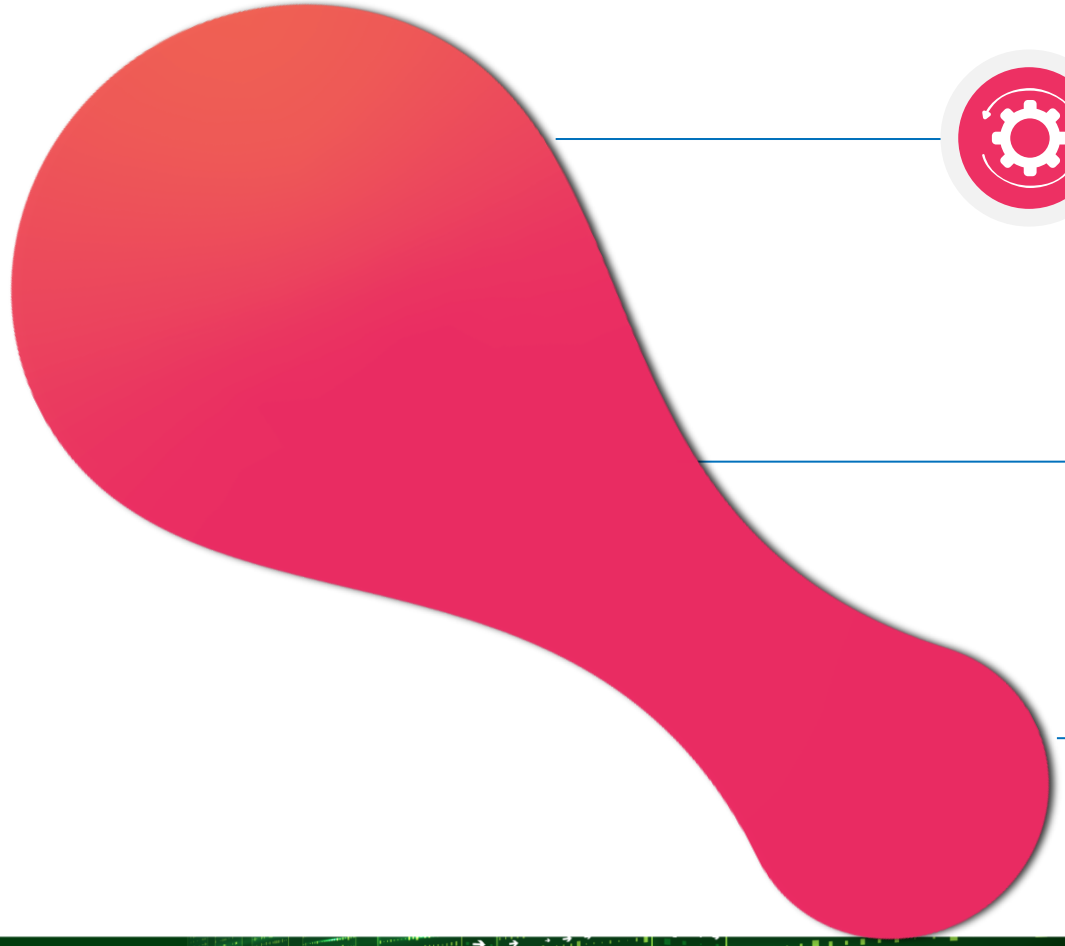
From Quantum Research to Deployable Networks

- Quantum communication technologies remain fragmented across protocols, components and systems
- Scaling beyond individual point-to-point links remains a core challenge
- Integration with classical telecom infrastructure is essential for real deployment
- Interoperability is required across hardware, software and network layers
- Large scale quantum communication networks must be validated in operational environments
- QSNP brings research, industry and telecom together to develop, integrate and deploy quantum communication technologies that protect data, including against future quantum computing threats
- Falqon® Systems contributes scalable MDI-QKD network architecture, system integration and deployment expertise

Bringing research, integration and deployment together at network scale.



What QSNP Does



DEVELOP

Advanced technology for quantum secure communication networks against the ever increasing power of computers and sophistication of algorithms (even for quantum computers).

INTEGRATE

Quantum cryptography technology at component, system and network levels, also into classical communication.

DEPLOY

Technology into Quantum-safe critical governmental infrastructures, private telecommunication market sector and future quantum internet.

EUROPEAN QUANTUM TECHNOLOGIES FUNDING OPPORTUNITIES

erc

MARIE CURIE ACTIONS

HORIZON EUROPE
2021-2027

RESEARCH BASED

HORIZON EUROPE

INFRASTRUCTURES

DIGITAL EUROPE

DIGITAL EUROPE
DIGITAL SKILLS, CYBERSECURITY, SUPERCOMPUTING

QUANTUM FLAGSHIP

Bring quantum **technologies** from the **lab** to the **market** and consolidate European scientific **leadership** in quantum research

- **FUNDAMENTAL R&D**
- **TECHNOLOGY SUPPLY**

ADVANCED DIGITAL SKILLS

Develop short term **training courses** and **Master programmes** in key capacity areas

QUANTUM COMMUNICATION INFRASTRUCTURE (EuroQCI)

Build and **deploy** in the next decade a certified secure pan-European end-to-end QCI for **cybersecurity services**

- **QKD INFRASTRUCTURE**
- **TESTING OF CROSS-BORDER QCI LINKS**

EuroHPC

QUANTUM COMPUTING INFRASTRUCTURE (EuroHPC)

Build and **deploy** an infrastructure for big data, artificial intelligence, high performance computing, among others

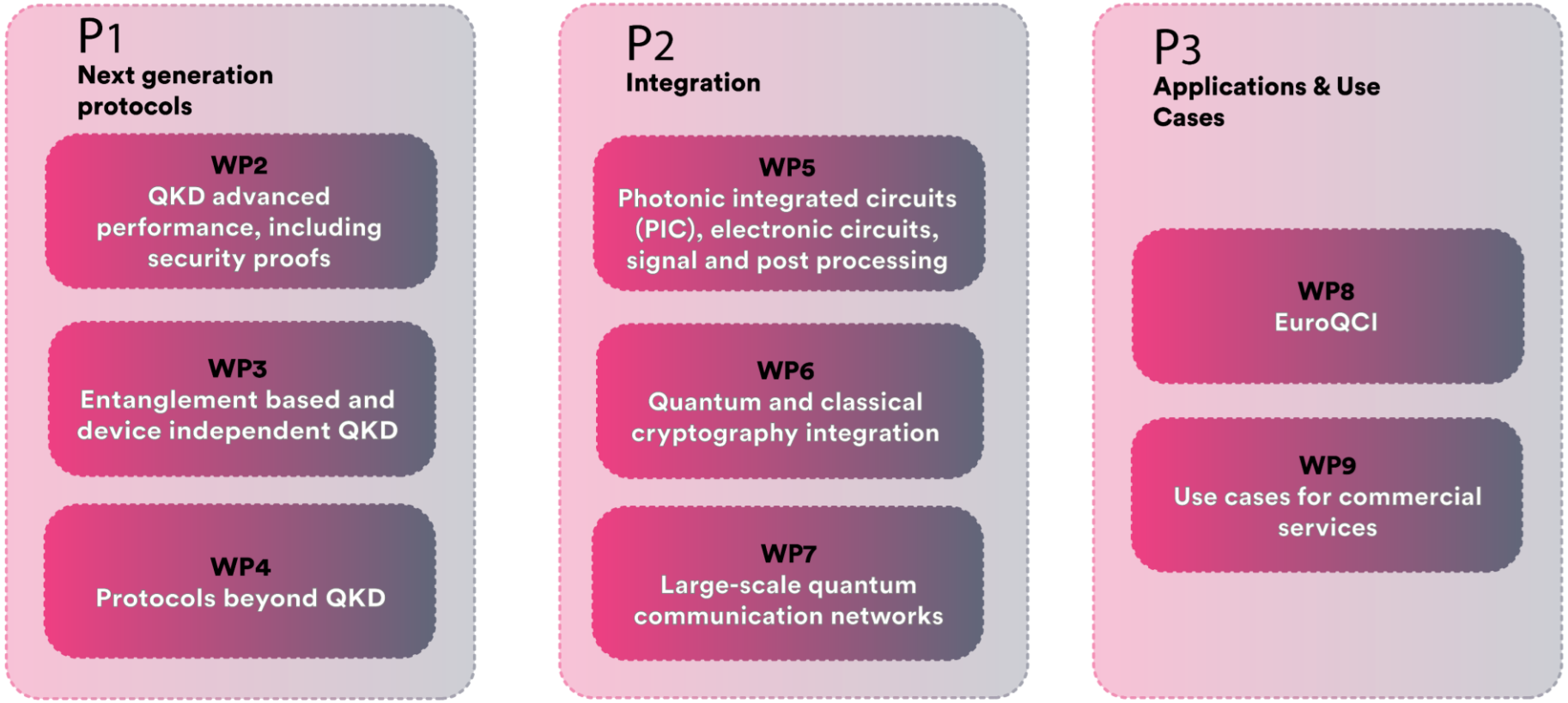
- **QT/HPC HYBRID**
- **QUANTUM SIMULATION/ COMPUTATION**

OCP FUTURE TECHNOLOGIES SYMPOSIUM | 2026

Scaling Innovation for the AI Era.

Pillars And Science & Technology Activities

Cryptography and Security, Networks, Systems & Components



Management (MGT)

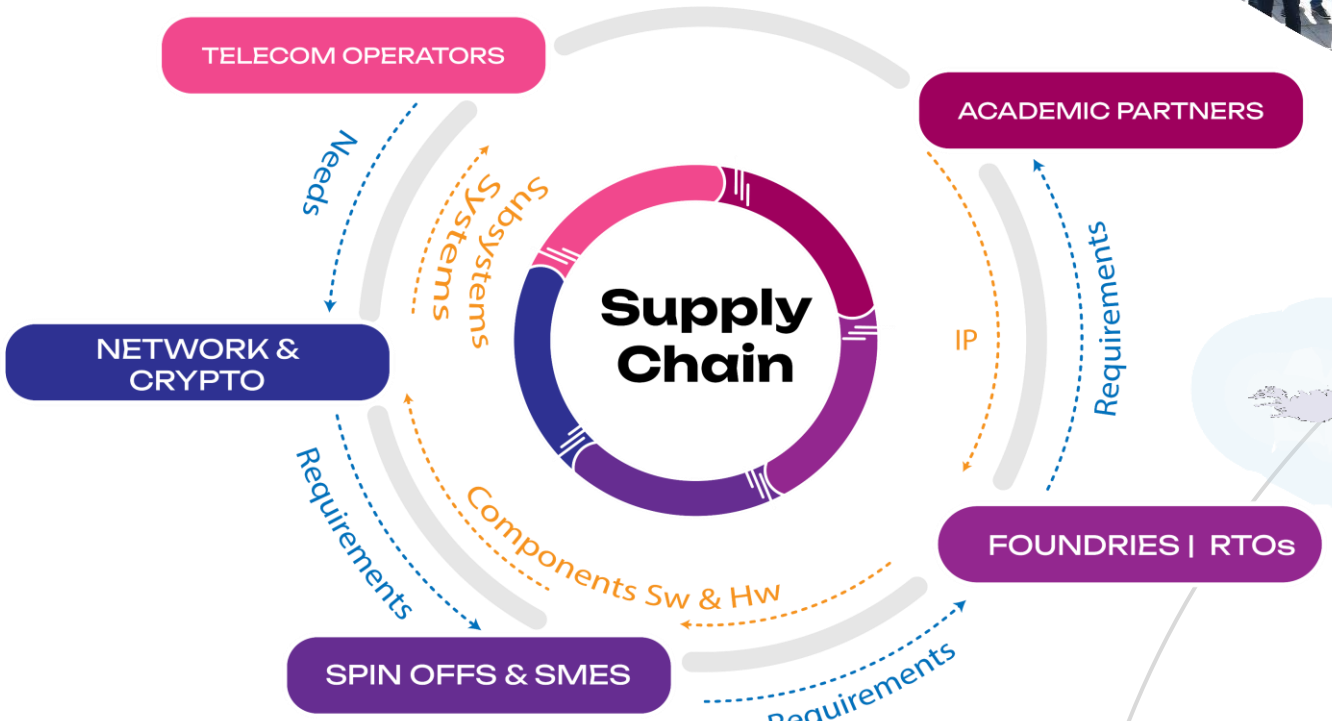
Ecosystem Building (EcoB)

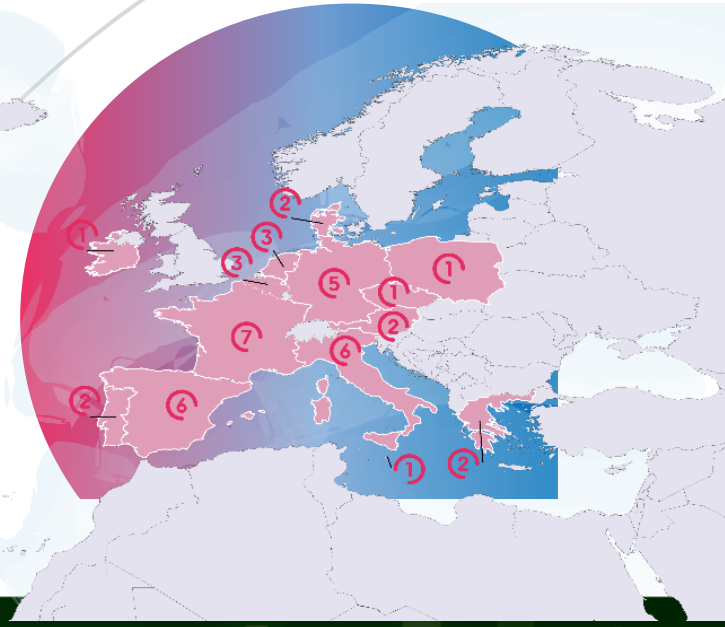
WP1
Project Management and Coordination

WP10
Dissemination, Exploitation & Ecosystem Building

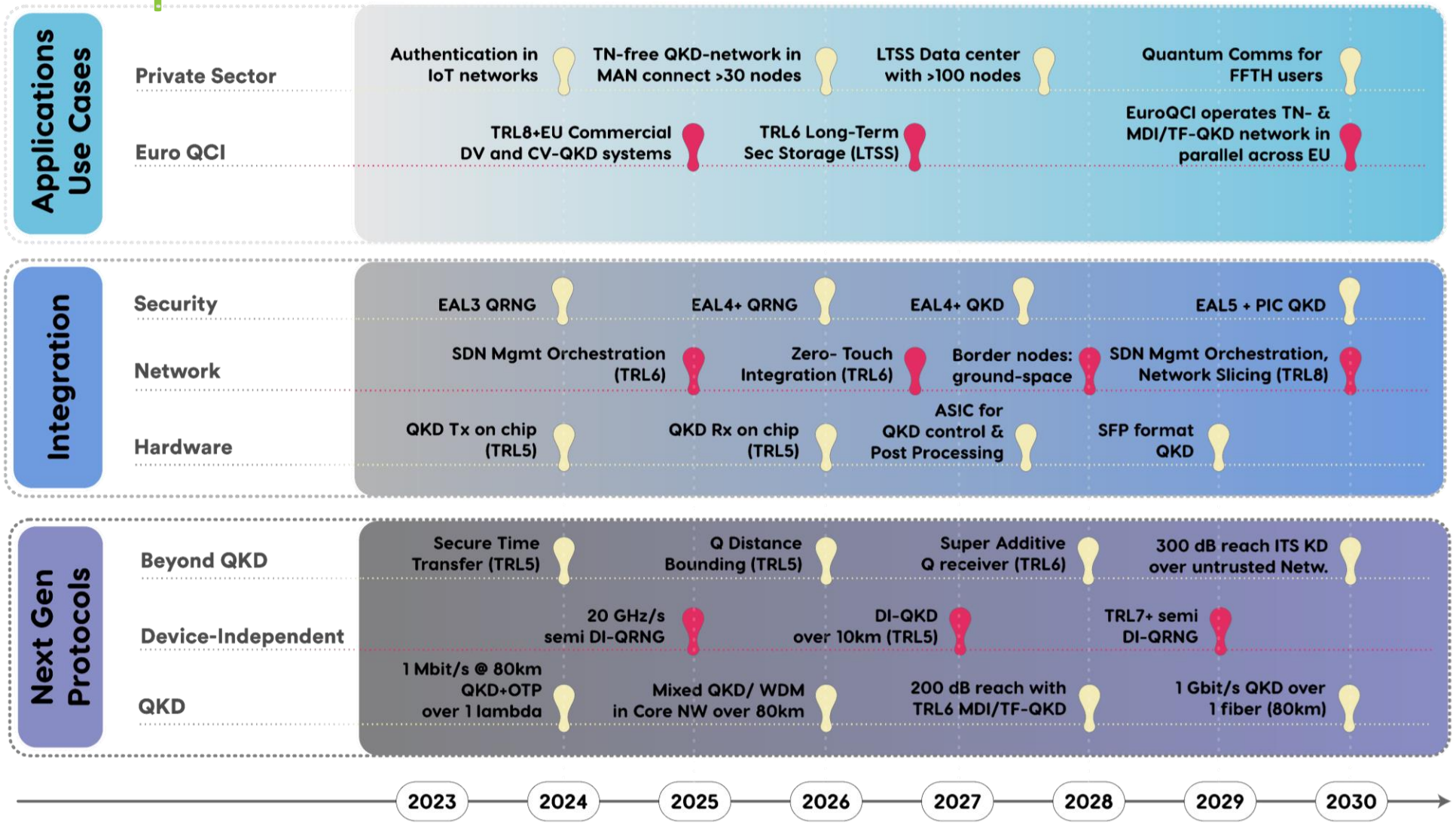
Meet The QSNP Team







Roadmap


Applications Use Cases
Private Sector
Euro QCI
Authentication in IoT networks
TN-free QKD-network in MAN connect >30 nodes
LTSS Data center with >100 nodes
Quantum Comms for FFTH users
TRL8+EU Commercial DV and CV-QKD systems
TRL6 Long-Term Sec Storage (LTSS)
EuroQCI operates TN- & MDI/TF-QKD network in parallel across EU
Integration
Security
Network
Hardware
EAL3 QRNG
EAL4+ QRNG
EAL4+ QKD
EAL5 + PIC QKD
SDN Mgmt Orchestration (TRL6)
Zero- Touch Integration (TRL6)
Border nodes: ground-space
SDN Mgmt Orchestration, Network Slicing (TRL8)
QKD Tx on chip (TRL5)
QKD Rx on chip (TRL5)
ASIC for QKD control & Post Processing
SFP format QKD
Next Gen Protocols
Beyond QKD
Device-Independent
QKD
Secure Time Transfer (TRL5)
Q Distance Bounding (TRL5)
Super Additive Q receiver (TRL6)
300 dB reach ITS KD over untrusted Netw.
20 GHz/s semi DI-QRNG
DI-QKD over 10km (TRL5)
TRL7+ semi DI-QRNG
1 Mbit/s @ 80km QKD+OTP over 1 lambda
Mixed QKD/ WDM in Core NW over 80km
200 dB reach with TRL6 MDI/TF-QKD
1 Gbit/s QKD over 1 fiber (80km)
2023
2024
2025
2026
2027
2028
2029
2030

Highlights

DEMONSTRATION OF QUANTUM-DIGITAL PAYMENTS

Peter Schiansky, Julia Kalb, Esther Sztatecsny, Marie-Christine Roehsner, Tobias Guggemos, Alessandro Trenti, Mathieu Bozzio & Philip Walther

Nature Communications volume 14, Article number: 3849 (2023)

QKD DEMO: A PROTOTYPICAL FULL-STACK MINI-EUROQCI

UPM, LuxQuanta, ThinkQuantum et al.

Stockholm (Sweden), Hannover (Germany), Barcelona (Spain)

- Secure transference of information over 100 kilometres via fire optic cable using quantum encryption.
- Successful integration of an O-band commercial QKD system over a testbed that replicates carrier-grade Fiber To The Home (FTTH) optical access network, emulating real-life deployments.
- Advances in the field of device-independent quantum information.



QSNP: Project Results (2023-2026)

QSNP start-ups and spin-offs

LUXQUANTA

Falqon Systems

Quside

ALEA

NEXTWORKS
HEADING THE FUTURE

ThinkQUANTUM

3 new spin-offs
Launched in 2024-2025

QUBO
quantum for everybody.

QoolNet

CELARE

PUBLICATIONS
Scientific papers 2023-2025
+200

TALENT
Talent attraction within research groups and companies
+100

ECOSYSTEM
3 spin-offs launched
3 in the pipeline for 2026
3

EVENTS & CONFERENCES
Attended events from QSNP speakers & representatives
+85

IP–Intellectual Property
3 patents
Up to 21 for 2026
3

Ecosystem

PUBLIC AUTHORITIES

European Commission co-funded projects like QSNP and EuroQCI to deploy a backbone QKD network within EU.

DEMONSTRATIONS

Live demos of the current state-of-the-art.

10

TELECOM PROVIDERS

- Telefónica: established the largest QKD network in Europe.
- Deutsche Telekom: key role in EuroQCI, coordinating PETRUS.
- Orange: coordinators of ParisQCI (French EuroQCI)

BUILT IN EUROPE. DESIGNED TO SCALE.

Falqon Systems

Scaling Quantum Secure Networks From Metro To Global Connectivity



Falqon® Systems is a Netherlands (Delft)-based leader in secure communications, enabling sovereign, scalable and operational Quantum Secure Networks (QSN) from metropolitan deployments to connectivity across borders for government, telecom, NREN and critical infrastructure.

Engineered In Europe



- Formerly Q*Bird, evolved into Falqon® Systems
- Founded in Delft, The Netherlands
- Designed, developed and manufactured in Europe
- Quantum research engineered into deployable MDI-QKD systems.

Networks Beyond Borders



- From metropolitan deployments to national and cross-border networks
- Integrated quantum systems, key management and orchestration built to scale
- Strengthening digital sovereignty and long-term cyber resilience

Proven In The Field



- 7+ European countries and counting across our growing international footprint
- 1419 kms of quantum network deployed including critical infrastructure at the Port of Rotterdam
- 50+ Falqon® network devices deployed

From a single secure link to scalable quantum secure infrastructure.

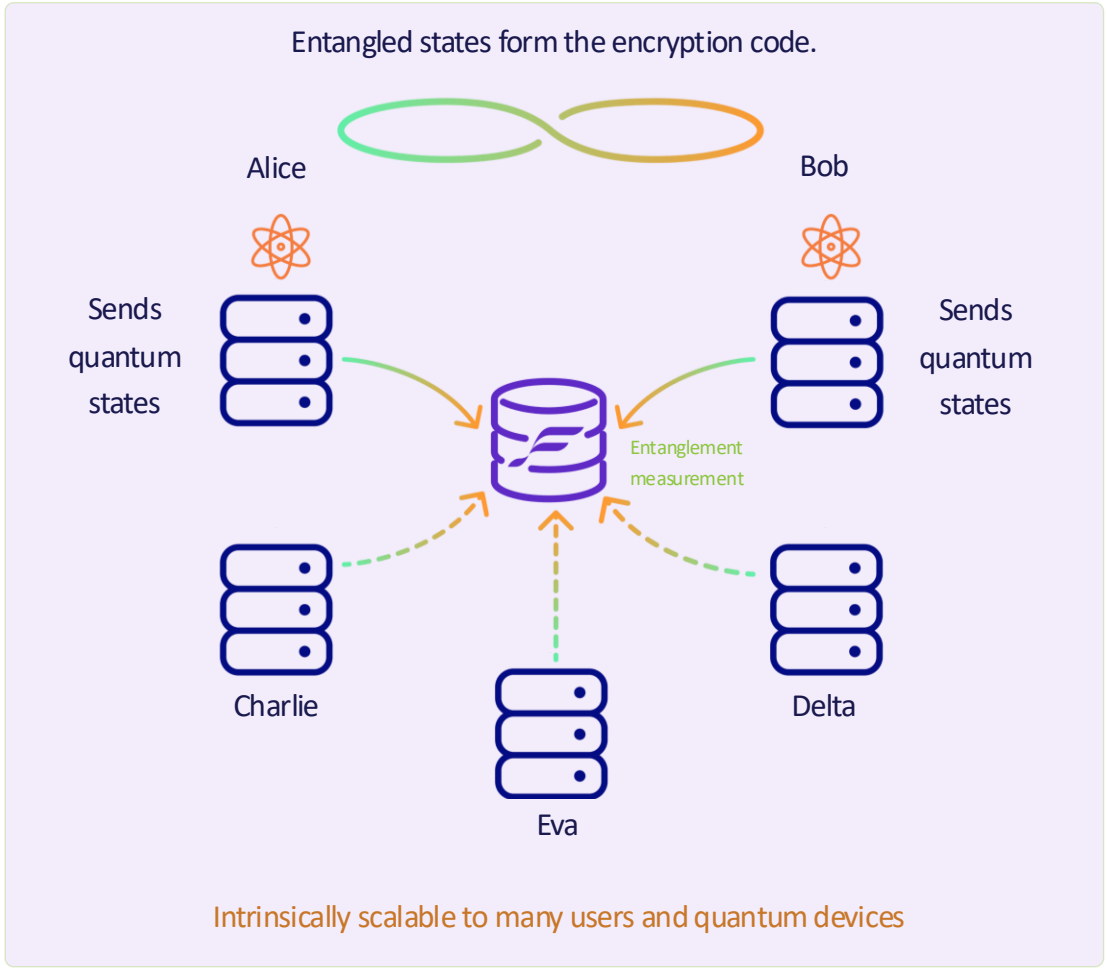
Falqon Systems

A Network-First Architecture For Scalable Quantum Secure Networks

More Scalable, More Resilient, More Secure.

- Falqon® Systems enables Entanglement Generation
- Uses the Measurement Device Independent MDI-QKD Protocol
- An Intrinsic Network Approach, beyond P2P
- Inherent, cost-effective scalability to many QKD Users, all without Trusted Nodes
- Resilient networks with redundancy and dynamic routing
- More secure with fundamental protection against side channel attacks
- Enables connecting quantum processors, quantum repeaters & quantum computers
- Future-proof security against quantum/AI computing threats

Not just QKD - A Quantum Secured Network



Falqon Systems

From Research to Operational Network Infrastructure

2021-2022
Proof of Concept



Deployed in data centers

2023-2024
Launch our MVP



First pilot in Port of Rotterdam

2024-2025
Scale Deployment



Shipping to NL, IE, BE, LU, AT, ES, DE

2026
Launch our Production



Grow customer base

Field experience drives product development, turning research breakthroughs into deployable network infrastructure.

Falqon Systems

Falqon® Systems - Turning Technology Into A Product Portfolio

MQT4000
QKD End Node



The qubit transmitter with the Embedded QKMS

MQX4000
Center Hub



The qubit receiver

MQS4000/MQS8000
Quantum Optics Switch



The qubit optical transport

Falqon® Domain Controller



Network management platform

Falqon® Key Manager

Key lifecycle management across distributed network environments

Deployment: *Within Falqon® QKD Nodes, as a standalone on-premise KMS device, on customer-provided servers or within customer-provided cloud environments.*



Falqon Systems

Making Quantum Security Work With Existing Infrastructure

Cisco Catalyst C8000 Series	Nokia 1830 PSD	Nokia 1830 SMS	Adtran FSP 3000	Juniper EX4400
				
IPSec	OTNSec	SNMP	OTNSec	MACsec

Quantum security designed to integrate with existing infrastructure, not replace it.

Falqon Systems

Proven in the Field: Falqon® Systems' Deployment Footprint

- **Integration** with Juniper, Adtran, Nokia & Cisco Line Encryption
- Network deployed in the **Port of Rotterdam**
- Networks deployed in **Industrial Data Centers**
- Network deployed in **Dutch Government Data Centers**
- Deployed **Spain's First Metropolitan-Scale network** in Madrid, with Multi-Node MDI QKD tech
- Deployed **Ireland's First Metropolitan-Scale Network** in Dublin, Multi-Node MDI QKD tech
- First **Cross-Border MDI QKD Quantum Network** deployment in Benelux together with Belnet - 150 km network



Falqon Systems

From Secure Links to Scalable Network Infrastructure

- **Built to integrate:** Works with existing network infrastructure.
- **Designed to scale:** From individual links to multi-user networks.
- **Proven in the field:** Deployments that drive continuous product development.

Falqon® Systems is building the infrastructure for Quantum Secure Networks, securing today's data center and telecom environments while enabling tomorrow's connected quantum infrastructure.

Join us in the quantum revolution!



Falqon Systems www.falqonsystems.com remon@falqonsystems.com FalqonSystems

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

